

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R61 responds to the 9 October 2026 advisory report on R59. The original Neural Bellman Operators paper gains root-free exact witnesses, matched native ablations, fixed-cover stability and original-optimum certification of unchanged returned policies. Earlier theory, applications and adverse evidence remain preserved.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R61*

We thank the referee for recognizing the exact witness and prospective-transcript results while identifying the further evidence needed to evaluate them as a numerical method for economics. The revision develops the original **Neural Bellman Operators** paper. It retains the title, nonlinear investment primitives, constructive Bellman-accuracy objective, trained continuation backend, acquired-policy implementation and broader applications. We have not replaced the paper by a differently titled screening note.

The controlling report is the owner-commissioned advisory review of R59 dated 9 October 2026, commit 8f56a3ae24ef3ce4383d0e9d1d757bd3ed7378c6, on manuscript 00412e6a4f43100996b324ea1ace097232c3fbbf. The R60 completed scientific execution and the separately frozen centered-Bellman execution supply subsequent materials that had not been integrated into the ordinary manuscript. R61 adds a full independent reconstruction, a new stability argument, original-optimum certification of every distinct returned two-date actor, and a fresh matched native two-by-two experiment. The advisory report is not represented as a decision of the Econometric Society.

The new sections are **Exact action recovery and the Bellman accuracy contract**, **Stable enclosures and certification of a returned policy**, and **Exact computation**

and the economic accuracy contract. The main article contains the statements and proofs; the supplement records the full service registry, ablations, failure conditions, source identities and numerical semantics. A unified introduction replaces the two successive introductory inputs while preserving those original files and the complete R59 documents. All inherited labeled results and adverse evidence remain available.

1.2. B1. *Separate the generic exclusion rule from the model-specific contribution*

The revision explicitly separates interval exclusion, exact activation reduction, the integrated objective's discrete curvature and the prospective policy contract. We do not claim the generic idea of discarding a candidate whose valid lower bound exceeds an incumbent upper bound. That principle is used to implement the original mathematical witness, including its smallest-index tie rule.

The new root-free theorem exploits the integrated NBO objective directly. On a scalar activation piece, the polynomial is $d_0 + d_1a + d_2a^2 + c_4a^4$ with $c_4 \geq 0$. Its second forward difference on a nonnegative lattice is nondecreasing, even if $d_2 < 0$. Exact rational sign comparisons locate the concave prefix and convex tail. At most four retained indices per nonempty piece supply the original smallest minimizer. The theorem therefore does not require one screening survivor, a unique optimum, a favorable growth constant or a root-isolation subroutine. A signed or flat fitted continuation is admitted under the stated polynomial structure.

The arithmetic-operation count, stored rational objects and bit-length dependence are given separately. The implementation is an exact C++ rational engine, not a floating root finder. The corresponding lossless replacement preserves the prior policy and stopping transcript under the same mathematical controller, while the fitted-witness transfer theorem retains the remaining acquisition, continuation, continuous-action and verification errors.

The multi-action extension has its own theorem rather than borrowing the scalar result. A finite mixture of independent box-uniform shocks has an exact signed-vertex ReLU expectation formula. Rational subdivision on a polytope-constrained action lattice preserves the original lexicographic optimum and terminates with an

exact fallback when a declared prefix budget is exhausted. The explicit 2^q shock-integration terms and worst-case product action-lattice size remain in its cost account. Sixteen fitted two-control queries per worker execute shared capacity and one or two shocks. This is a substantive non-scalar exact-witness extension, distinct from a complete multivariate-control welfare experiment.

1.3. B2. Stress the mechanism beyond the favorable one-survivor regime

The integrated R60 catalogue has ninety rational scalar instances per worker: three fixed seeds, widths eight and thirty-two, lattice cardinalities thirty-three, 129 and 513, and five regimes. Mixed features, crossing activations, large signed cancellation, adjacent exact ties and flat objectives are all retained. Every result is compared with exhaustive Fraction minimization of the original objective and original tie rule.

A fixed subset receives seven additional shuffled repetitions on each worker. Coarse endpoint diagnostics use eight and twenty-four fractional bits, and the guard catalogue reaches lattices of 2050 and 4097 points. Screening and adaptive fallbacks are reported rather than treated as implementation failures to be removed. The main theorem does not rely on a one-survivor observation, and the new finite-difference construction handles flatness and ties without forcing separation.

The supplement explains precisely what was varied: rounding an exact bound outward to a coarser endpoint grid tests screening conservatism, not a different internal floating-point processor precision. All per-instance outputs, objective payloads, operation counts, survivor counts and failure-prefix records are preserved. The R61 audit checks 2,408 scalar solver records, including repeated and guard cases, without treating those repetitions as additional learned models.

1.4. B3. Isolate reduction, screening, language and algorithm effects

The eight-method experiment includes unreduced and reduced symbolic search, vector enclosure with exhaustive exact comparison, unreduced and reduced screens, native exhaustive evaluation, native finite-difference search, and adaptive subdivision. These comparisons expose useful mechanisms, but source inspection identified

a remaining confound: the original native exhaustive mode was unreduced, whereas the native piecewise mode first reduced activation regions.

R61 therefore adds a new, prespecified native two-by-two experiment. Its four cells cross reduction off/on with exhaustive/finite-difference search. Every cell uses the same C++ arbitrary-size rational engine, original objective, lattice, smallest-index rule and exact original-value postcheck. Reduction work is included wherever reduction is performed. The ninety original rational fixtures and seven-repeat subset are fixed before the fresh timing observations; no favorable instance is selected using those observations. Two newly allocated worker jobs execute all cells. Batch sizes one, eight and thirty-two are also evaluated for the same complete ninety-query catalogue.

This comparison identifies the search effect at a common reduced representation. At lattice cardinality 513, the median recorded saving is approximately 61–73 percent across the width/worker groups, and every instance in those groups is faster. At width thirty-two and lattice cardinality thirty-three, the medians are approximately minus eighteen and minus nineteen percent: native exhaustive search is faster. The revision therefore reports the actual crossover in the tested catalogue instead of claiming that the piecewise method always wins. All four repeated-query medians, full ranges, random orders, batch bytes and exact answer arrays remain available. The audit independently verifies 3,160 returned exact answers; that count is not an economic sample size.

1.5. B4. *Timing repetitions, hosts and uncertainty*

The stress subset has seven shuffled repetitions per regime and worker. The fresh native factorial uses independently shuffled within-instance method orders and records each timing, rather than only a selected median. The prospective execution has two hosted-worker repetitions of the full seed/task/target/method catalogue, with immutable parent checkpoints and complete child-return records.

The paper distinguishes mathematical designs from execution repetitions. Identical fitted objects and path streams on two worker allocations do not create independent policy-cost observations. Likewise, separately allocated hosted jobs do

not prove independent physical hardware, processor-frequency control or cross-architecture generality. CPU affinity and processor/compiler/runtime metadata are recorded. The reported hardware-counter probe did not produce counter values under the available permissions; those values are not invented.

Within-worker paired descriptive contrasts and complete timing ranges are reported. Query timing, batched query timing, warm full-service time and cold compilation-inclusive time have different boundaries. None is substituted for another. The observed negative small-lattice savings are part of the timing result, not discarded outliers.

1.6. B5. *Connect exact fitted actions to actual policy and economic outcomes*

Every complete service retains its fitted critics, common finite-menu actions, exact extra witness, selected whole-cell action, true reference-advantage endpoints and final actor identity. The nested-candidate design compares the selected policy before and after the extra witness under the same verifier and shared actions. The independent reconstruction checks all acquired cell-date decisions and identifies additional-witness changes across the distinct verification objects, not only the total number of candidate evaluations.

The exact native and screened ReLU implementations have the same fitted parameters, witnesses, complete policies, certificates, stopping stages, path counts and confidence intervals. Their resource comparison is therefore at an unchanged economic answer. Different generators are compared through a fresh final-policy cost stream, with native-ReLU-minus-common and native-ReLU-minus-quadratic intervals shown for every task, target and seed. Policy identities are exact zero contrasts; other intervals containing zero remain unresolved. A smaller local upper advantage is not equated with a statistically identified aggregate welfare gain.

The new fixed-policy Bellman calculation adds a further link. It certifies the actors that were actually returned, using their original discontinuous acquisition partitions, rather than assigning a separate comparator's bound to a neural policy. No action, training seed or initial distribution is replaced during that calculation.

1.7. B6. Compare complete NBO services with strong conventional alternatives

The new completed catalogue comprises ninety-six services: four modes, three dimension/horizon pairs, two seeds, two economic targets and two worker repetitions. The modes are common-only, native quadratic, screened ReLU and native ReLU. The original tree, tensor, compiled-witness and nonuniform conventional comparisons remain in the preceding evidence. Common-only is a genuine competing service when verification of a fixed action family suffices; its low cost is not excluded because it has no fitted continuation.

Each method pays for its own training when applicable, all executed construction attempts, exact search, whole-cell verification, sequential inference, durable scientific records, process log and first parent receipt. Cold native use additionally pays the measured compilation expense; warm use reports compilation separately. Eighty services attain their declared targets and sixteen exhaust the declared budget. All ninety-six returns, unsuccessful targets and exact final cost endpoints appear in the registry.

The service table reports each method at the same target, and the separate paired table reports actual native-ReLU cost differences relative to the conventional policies. The revision does not infer a conventional-method victory from a neural certificate alone, or a neural-method victory from a cheaper exact implementation of the same neural policy. It supplies the complete empirical comparison on which that judgment can be made. Universal neural dominance is not assumed as a premise of the NBO theory.

1.8. B7. Establish accuracy against the original Bellman optimum

The original constructive objective is retained and now has an executed direct accuracy comparison. Whole-state lower intervals cover all continuous feasible actions. Upper candidates are feasible throughout their acquired cells. The original continuous innovation laws are enclosed rather than replaced by midpoint transitions. The uncentered bracket remains in the record, including its relatively loose final result.

The common-reference construction transports the lower Bellman excess and the implemented-policy upper excess relative to the same zero-policy continuation. Their difference cancels that reference at the same state. The centered ladder attains tolerances one-half, one-quarter, one-eighth and one-sixteenth. At $N = 128$, its maximum datewise original-optimum gap is approximately 0.039787; one-thirty-second is not attained on the declared ladder. The cumulative centered construction work includes every earlier rung and serialized endpoint array.

That bracket initially belongs to its own constructed policy. R61 separately revalidates all thirty-two actually returned two-date service records, which correspond to five complete policy-plus-partition identities. The protocol fixes $N = 128$, $A = 64$, $q = 32$ before the new actor calculations. It includes every identity irrespective of generator, target or observed cost. Three actors have maximum datewise gaps approximately 0.067777–0.071752, attaining one-eighth but not one-sixteenth; two retain gaps approximately 0.303297. Thus the accuracy result for returned neural and conventional actors is stated under its own numerical bound, rather than inheriting the separate Bellman comparator’s tighter value.

Both clipped and unclipped upper recursions are retained. Intersection with zero is used only after independent reconstruction of the original reference-advantage gate proves global nonworsening relative to zero investment. In this catalogue the maximum gaps coincide with the unclipped calculation. Complete leaf-range enclosures include actor discontinuities and closed boundaries. Verification does not change the policy and creates no policy-cost sampling observation.

We also strengthened the proof of refinement consistency. A valid endpoint at the true continuation is insufficient by itself to imply stable propagation at approximate continuations. The revised statement explicitly requires fixed-cover nonexpansivity with separately budgeted arithmetic error. The new lemma proves this property for the actual min/max, nonnegative averaging and action-minimization maps. A counterexample explains why inclusion alone would be insufficient. The runtime bracket remains valid independently of the asymptotic consistency clause, and arbitrarily small tolerances require refinement of every binding discretization, including the action quantum when relevant.

1.9. B8. Rational bits, memory, batching and fallback complexity

The new action theorem gives an explicit rational-operation count, storage count and conservative operand-length bound. The rational-arithmetic cost multiplies that operation count; arbitrary-size arithmetic is not assigned unit cost. The implementation's measured operand diagnostic is named accurately: it observes designated inputs, coefficients and evaluations, not every temporary integer inside the library. The proof's operand bound and the observed diagnostic are not conflated.

For the scalar piecewise objective, finite differences replace root isolation and have logarithmic dependence on lattice cardinality per piece, but direct piece construction and original objective evaluation still incur feature-dependent work. At short lattices that overhead can make exhaustive evaluation faster, as the matched native experiment demonstrates. The multivariate subdivision theorem retains the worst-case product lattice size and the cost of valid bounds and polytope feasibility. The box-shock integration formula retains its exponential term count in the number of independent components.

The supplemental tables retain active features, exact comparisons, nodes, fallback prefixes, recorded interval-array bytes and operand diagnostics. Complete services separately report Python and child peak resident memory. The fresh batch experiment records request and response sizes, subprocess counts and all exact answers. Compilation, query reduction, endpoint construction, exact comparison and durable return therefore have visible accounting boundaries rather than being hidden inside an asymptotic label.

1.10. B9. Economic interpretation, replacement fees and numerical units

The numerical units remain those of the original normalized investment model. No installation-price estimate, electricity tariff, processor-frequency estimate or external welfare calibration is inferred from hosted execution time. Instead, the manuscript gives the full symbolic decision account in those units.

For identical returned policies, an implementation change affects computation cost. At work price λ , setup difference $k_1 - k_0$, per-use work difference $w_1 - w_0$, use count H and additional implementation fee I , the new-minus-old total cost is

$\lambda[H(w_1 - w_0) + (k_1 - k_0)] + I$. This yields the entire break-even use-count account whenever a measured per-use saving exists, while a negative saving remains a negative result.

For different policies, a gain interval $[l, u]$, replacement fee τ and incremental work charge $\lambda\Delta w$ give net benefit $[l - \tau - \lambda\Delta w, u - \tau - \lambda\Delta w]$ on the same coverage event. The cases of supported replacement, ruled-out benefit and unresolved benefit are stated separately. New fixed-policy Bellman bounds supply a different accuracy contract and have their own paid verification cost; they are not treated as a reduction in an earlier measured stopping time.

1.11. *B10. Exposition, reading order and preservation*

The active introduction is rewritten as one argument from continuation construction to an exact fitted witness, an acquired-policy certificate and an economic return. It incorporates the earlier substantive positioning without successive revision-preface paragraphs. The new mathematical statements and complete proofs appear in the article, followed by a study organized around complete services, implementation mechanisms and original-optimum accuracy. Detailed registries and numerical semantics are in the supplement. The original title and topic remain unchanged.

All inherited theorem labels are checked for preservation. The original R59 introductory files and full ordinary sources are retained, as are complete companions covering recursive utility, temporal selves, diffusions, games and earlier numerical developments under their original hypotheses. Failed prospective targets, unfavorable conventional rankings, unsuccessful refinement rungs and the first recording failure remain accessible. This organization does not replace the NBO program by a narrower renamed paper, and it does not remove inconvenient evidence to create an apparent improvement.

1.12. *Verification and submission package*

The completed primary R61 reconstruction re-evaluates exact scalar and multi-action witnesses, every service's whole-cell certificate, all unique original-law path calculations, stopping decisions and the uncentered and centered Bellman ladders.

1 It verifies 1,128 interval records and reconstructs 1,896,448 path rows under the 1
2 original streams; these are not new independent samples. Fixed-policy bounds are 2
3 executed from their frozen protocol and then independently replayed. The new 3
4 native factorial is a fresh implementation-work experiment with all exact answers 4
5 independently checked. Original clocks and policies remain unchanged. 5

6 The seven-document publication package contains the active article, technical 6
7 supplement, response, complete article and proof companion, and retained develop- 7
8 ment article and proof companion. The publication gate checks source freezes and 8
9 preservation, runs inherited and new regressions, compiles the ordinary sources, 9
10 checks citations and typography, and rebuilds from a clean archive. Final files and 10
11 the remote review-ready revision are bound by the delivery manifest. Verification 11
12 and compilation support the submitted derivations and calculations; they are not 12
13 described as an external mathematical or editorial acceptance. 13

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